CAPSEL / INCA

Implementation Guide

GitHub-style repository architecture,
module responsibilities, data pipeline,
training loop design, and implementation order

Nishant Kumar · PhD Programme · May 2026
Supersedes: Phase0/ + Phase1/ layout

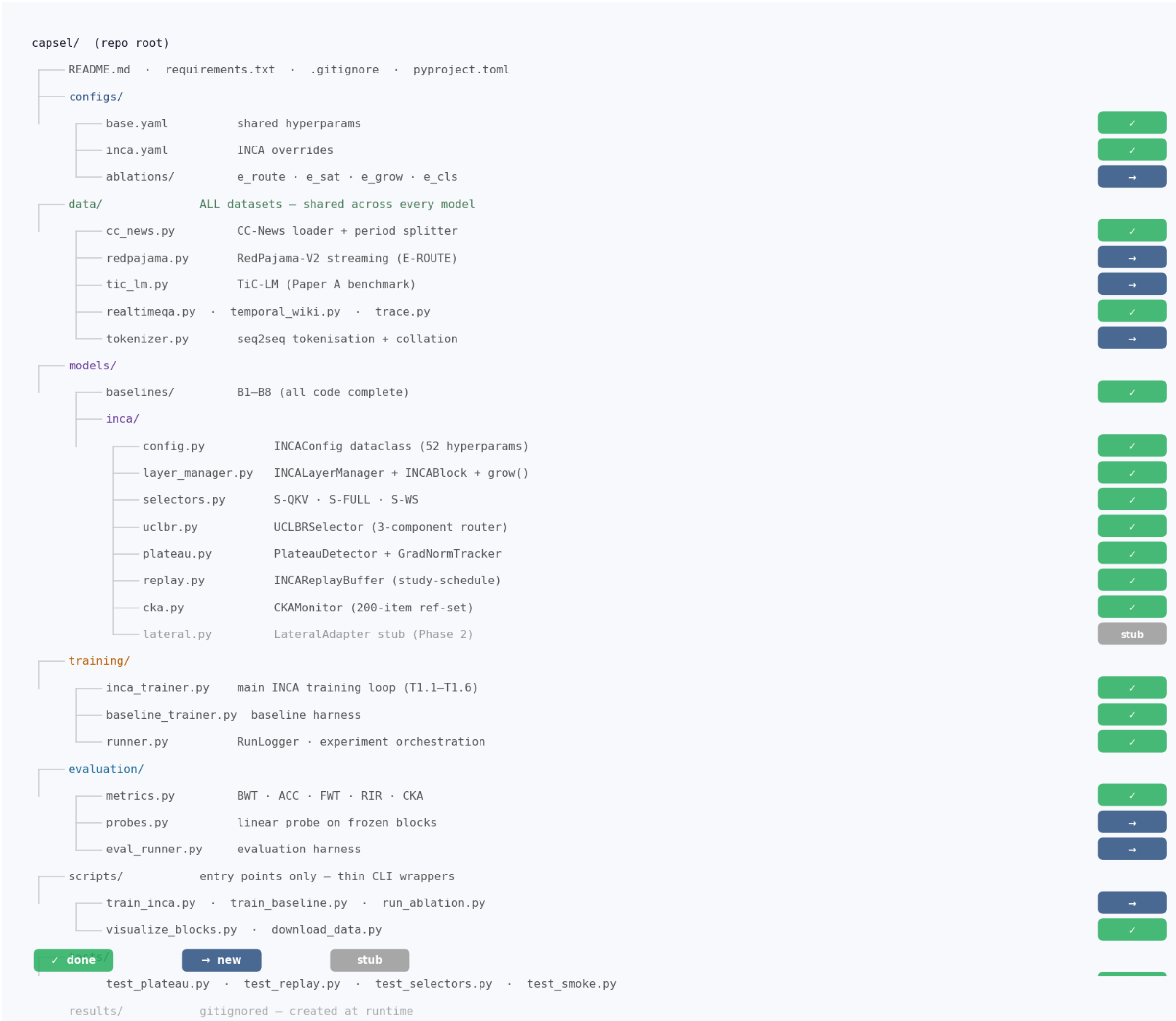
1. Why Restructure

The old layout split code by research phase (Phase0/, Phase1/) rather than by function. This caused three practical problems:

- **Dataset code was scattered.** CC-News downloads lived in Phase0/data/, but Phase 1 training needed the same data. No shared data module existed.
- **Common utilities were duplicated.** metrics.py, config.py, harness.py were imported with fragile relative-path hacks across phase boundaries.
- **No clean entry point.** Running any experiment required knowing which phase it lived in and manually setting sys.path. Uninviting for collaborators.

The new layout organises code by **what it does**, not **when it was written**. Dataset code is shared. Models, training, and evaluation are each a clean package. All experiments are launched from scripts/ with a single config flag.

2. Repository Layout



Green ✓ **done** = code already exists (migrated from Phase0/Phase1). Blue → **new** = needs to be written. Grey **stub** = placeholder for a future phase.

3. Module Responsibility Map

3.1 data/

File	Responsibility	Status
cc_news.py	Load vblagoje/cc_news, assign half-year periods, subsample 20K/period, seq2seq completion split (prefix→s	→ write
redpajama.py	Stream togethercomputer/RedPajama-Data-V2 by CC snapshot. 6 snapshots = 6 periods. Used for E-ROUT	→ write
tic_lm.py	Wrap Apple's TiC-LM pipeline (114 monthly CC dumps). Used only for Paper A final benchmark.	stub
realtimeqa.py / temporal_wiki.py	Existing Phase0/data/ loaders — copy verbatim.	✓ copy
tokenizer.py	FLAN-T5 tokenisation: 'complete: ' prefix, max_len=256 in+out, pad→-100 in labels. Returns tokenized Datas	→ write

3.2 models/inca/

File	Responsibility	Status
config.py	INCAConfig(Phase0Config): 52 hyperparams as a dataclass. YAML-loadable via OmegaConf. Single source	✓
layer_manager.py	INCALayerManager + INCABlock. Sequential chain with embedding skip. freeze_and_grow(), trainable_para	✓
selectors.py	EmbeddingQuerySelector (S-QKV default), CrossAttentionSelector (S-FULL), WeightedSumSelector (S-WS)	✓
uclbr.py	UCLBRSelector: Read-ME pre-gate + DeepSeek load-balance bias (online, no aux loss) + uncertainty-calibra	✓
plateau.py	PlateauDetector (multi-signal consensus: RIR, grad-norm, CKA, loss slope) + GradNormTracker (EMA). Emit	✓
replay.py	INCAReplayBuffer: per-block, cleared at freeze. Phase A uniform / Phase B study-schedule (70/20/10 hard/e	✓
cka.py	CKAMonitor: unbiased HSIC CKA on 200-item ref-set. Cached at period start, polled every k_eval steps.	✓
lateral.py	LateralAdapter placeholder — rank-r adapters from frozen→trainable blocks. Phase 2 implementation target.	stub

3.3 models/baselines/

File	Baseline	Status
finetune.py	B1 — naive sequential fine-tuning (forgetting floor)	✓
replay.py	B2 — replay-only (no architecture change)	✓
ewc.py	B3 — EWC regularisation	✓
l2p.py	B4 — Learning to Prompt (frozen backbone)	✓
lora_moe.py	B5 — LoRA-MoE (modular routing, no block growth)	✓
llama_pro.py	B6 — LLaMA-Pro (scheduled vertical growth — to beat)	✓
pnn.py	B7 — Progressive Neural Networks (lateral, no forgetting)	✓
block_stack.py	B8 — BlockStack variant (INCA internal reference)	✓

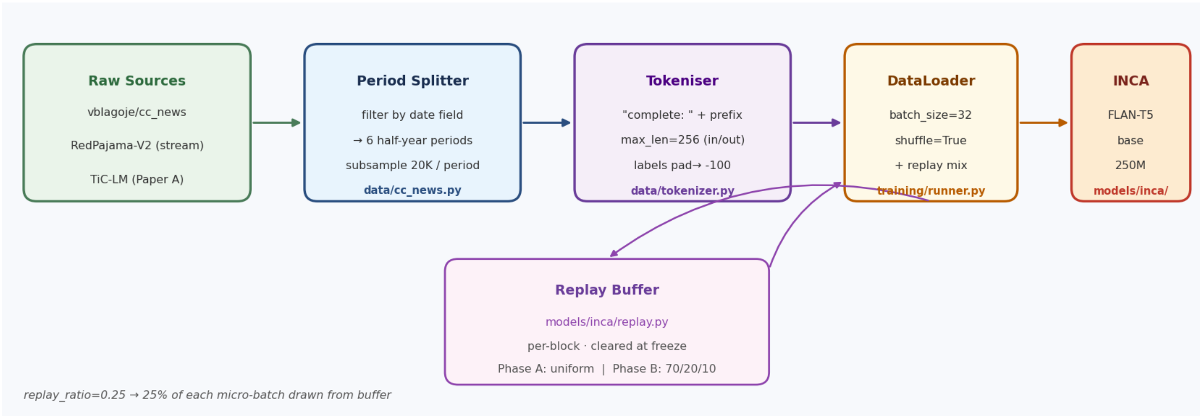
3.4 training/

File	Responsibility	Status
inca_trainer.py	Main INCA training loop. Period stream, micro-batch construction (stream + replay mix), loss computation, pla	✓ migrate
baseline_trainer.py	Generic continual learning harness for all 8 baselines. Shared period loop, metric collection, checkpointing.	✓ migrate
runner.py	RunLogger: timestamped output directory, loss_curve.csv, training.log, config_snapshot.json. Wipes old run c	✓ migrate

3.5 evaluation/

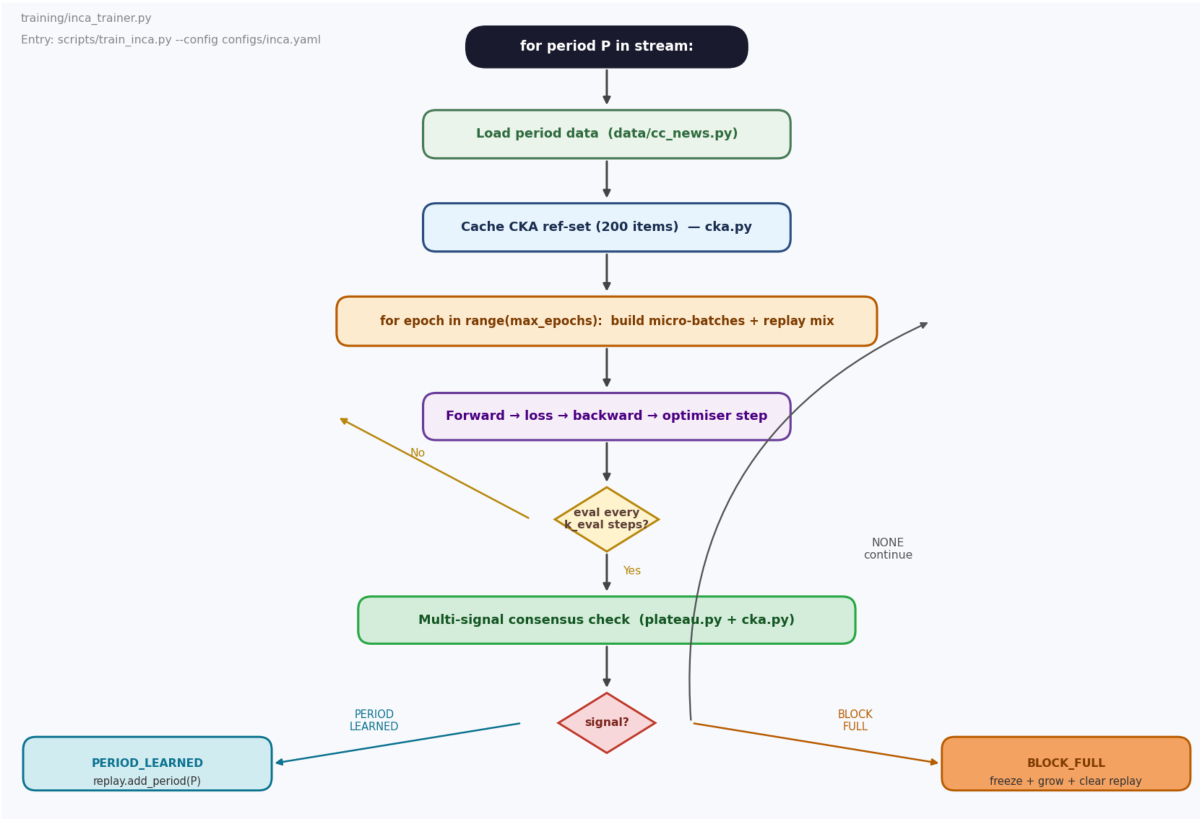
File	Responsibility	Status
metrics.py	BWT, ACC, FWT, RIR, per-block CKA, Fisher info, probe accuracy. Consolidated from Phase0/common/metr	✓ migrate
probes.py	Linear probe on each frozen block's mean-pooled representation. Supports E-CLS2 (monotone accuracy dec	→ write
eval_runner.py	Evaluation harness: load checkpoint → run on all periods → compute full metric suite → write JSON report.	→ write

4. Data Pipeline



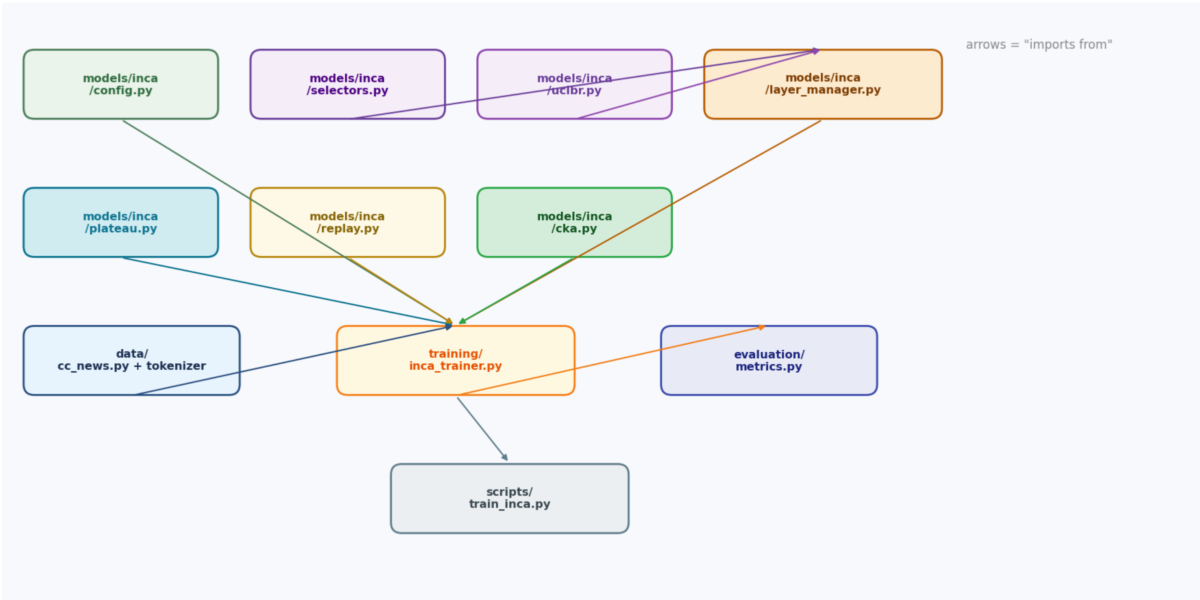
The **replay mix** in the DataLoader blends 75% stream items with 25% replay buffer items per micro-batch (controlled by replay_ratio=0.25 in INCAConfig). The replay buffer is phase-aware: Phase A (epoch < N_revise=3) samples uniformly; Phase B uses the 70/20/10 study schedule.

5. Training Loop Flow



The consensus check fires every k_eval optimiser steps. Any two of {RIR, grad-norm decay, CKA saturation, loss plateau} must agree before a signal is emitted. The grokking guard prevents BLOCK_FULL from firing before min_epochs_before_grow=2 full epochs.

6. Module Dependency Graph



The dependency graph is acyclic and layered. models/inca/ components have no knowledge of the training loop — they are pure PyTorch modules. The trainer imports everything; scripts import only the trainer.

7. Migration Map — Old → New

Old path	New path	Action
Phase1/src/inca_config.py	models/inca/config.py	move + rename
Phase1/src/inca_layer_manager.py	models/inca/layer_manager.py	move + rename
Phase1/src/inca_selectors.py	models/inca/selectors.py	move + rename
Phase1/src/inca_uclbr.py	models/inca/uclbr.py	move + rename
Phase1/src/inca_plateau.py	models/inca/plateau.py	move + rename
Phase1/src/inca_replay.py	models/inca/replay.py	move + rename
Phase1/src/inca_cka.py	models/inca/cka.py	move + rename
Phase1/src/train_inca_v3.py	training/inca_trainer.py	move + rename
Phase1/scripts/visualize_inca.py	scripts/visualize_blocks.py	move + rename
Phase1/tests/test_plateau.py	tests/test_plateau.py	move
Phase1/tests/test_replay_schedule.py	tests/test_replay.py	move + rename
Phase0/common/config.py	models/baselines/base_config.py	move + rename
Phase0/common/metrics.py	evaluation/metrics.py	move
Phase0/common/harness.py	training/baseline_trainer.py	move + rename
Phase0/common/runner.py	training/runner.py	move
Phase0/baselines/b*.py	models/baselines/*.py	move + strip prefix
Phase0/data/download_cc_news.py	data/cc_news.py	merge + extend
Phase0/data/download_*.py	data/*.py	move + rename
Phase0/configs/base.yaml	configs/base.yaml	move
Phase1/configs/inca_v2_smoke.yaml	configs/inca.yaml	merge

8. Config Hierarchy

All hyperparameters live in YAML configs loaded into the INCAConfig dataclass via OmegaConf. Three levels of override:

```
configs/
base.yaml ← learning_rate, batch_size, seed, max_epochs
inca.yaml ← inherits base; adds INCA-specific params
ablations/
e_route.yaml ← selector: [embedding_query, uclbr, cross_attention, weighted_sum]
e_sat.yaml ← rir_threshold sweep
e_grow.yaml ← layers_per_block sweep
```

Example YAML for E-ROUTE ablation:

```
# configs/ablations/e_route.yaml
defaults:
- /configs/inca # inherit all INCA defaults

sweep:
selector: [embedding_query, uclbr, cross_attention, weighted_sum]
seed: [42, 123, 999]

# Fixed for all runs:
dataset: cc_news
n_periods: 6
items_per_period: 20000
```

8.1 CLI entry points

```
# Single INCA run
python scripts/train_inca.py --config configs/inca.yaml

# Single baseline
python scripts/train_baseline.py --baseline b6 --config configs/base.yaml

# Ablation sweep (launches grid via subprocess or SLURM)
python scripts/run_ablation.py --config configs/ablations/e_route.yaml

# Evaluate a checkpoint
python scripts/eval_run.py --checkpoint results/inca_run_001/inca_final.pt

# Visualise block chain
python scripts/visualize_blocks.py --checkpoint results/inca_run_001/inca_final.pt
```

9. Implementation Order

Files marked ✓ exist and only need path/import fixes. Files marked → need to be written. Order below minimises blocked work — each layer can be tested before the next.

Order	File	What to write	Unblocked by	Test
1	data/tokenizer.py	FLAN-T5 tokenise fn, completion split, DataLoader factory with replay mix	Nothing — pure HF	test_tokenizer.py — check shapes
2	data/cc_news.py	Period splitter using date field, subsample 20K, call tokenizer.py	#1	test_data.py — period counts
3	models/inca/config.py	Copy Phase1 inca_config.py verbatim; update import to not need sys.path hack	Nothing	python -c 'from models.inca.config import INCAC
4	models/inca/*.py	Copy all 6 inca_*.py files; rename to drop inca_ prefix; fix imports (from .config import INCAConfig)	#3	pytest tests/test_phase1.py tests/test_replay.py
5	training/runner.py	Copy Phase0/common/runner.py verbatim; update imports	#3	Instantiate RunLogger — check dir created
6	training/inca_trainer.py	Copy train_inca_v3.py; rewire imports to new package paths; wire data/ for period loading	#1 #2 #4 #5	smoke run: 2 periods, 100 steps each
7	evaluation/metrics.py	Copy Phase0/common/metrics.py verbatim	Nothing	test_metrics.py — BWT formula
8	scripts/train_inca.py	Thin wrapper: parse --config, build INCAConfig from YAML, call trainer	#3 #6	python scripts/train_inca.py --config configs/inca.
9	scripts/train_baseline.py	Thin wrapper for baseline harness	#7	python scripts/train_baseline.py --baseline b1 --d
10	evaluation/probes.py	Linear probe on mean-pooled frozen block reps	#4 #7	test_probes.py — accuracy not random
11	data/redpajama.py	Streaming loader for RedPajama-V2 6-snapshot period design	#1	stream 100 items — check fields

10. Test Coverage Plan

Test file	What it covers	Priority
tests/test_plateau.py	PlateauDetector: RIR, grad-norm EMA, CKA threshold, grokking guard, consensus any-2 rule	P0 — existing, 17 tests
tests/test_replay.py	INCAReplayBuffer: add/clear lifecycle, Phase A uniform sampling, Phase B study-schedule proportions, buff	P0 — Existing, 13 tests
tests/test_selectors.py	EmbeddingQuerySelector: per-block separate attention (not concat), output shape, weight sum=1. UCLBRS	P0 — write
tests/test_layer_manager.py	freeze_and_grow: block count increases, frozen params have requires_grad=False, new block is warm-start	P0 — write
tests/test_data.py	cc_news.py: period count=6, items per period ≤ 20K, input_text starts with 'Complete: ', target_text non-empt	P1 — write
tests/test_smoke.py	Full 2-period INCA smoke run on 50 items/period. Checks: model runs, loss decreases, no RuntimeError, ch	P0 — existing, update imports
tests/test_metrics.py	BWT formula (negative = forgetting), ACC monotone with more periods, RIR→0 at plateau, CKA∈[0,1]	P1 — write

11. requirements.txt

```
# Core ML
torch>=2.2.0
transformers>=4.40.0
datasets>=2.18.0
accelerate>=0.28.0
peft>=0.10.0

# Config
omegaconf>=2.3.0
pyyaml>=6.0

# Numerics
numpy>=1.26.0
scipy>=1.12.0

# Logging / utils
tqdm>=4.66.0
tensorboard>=2.16.0
pandas>=2.2.0

# Visualisation (optional — for scripts/visualize_blocks.py)
matplotlib>=3.8.0
seaborn>=0.13.0

# Tests
pytest>=8.0.0
```

11.1 pyproject.toml snippet

```
[tool.pytest.ini_options]
testpaths = ["tests"]
addopts = "-v --tb=short"

[tool.black]
line-length = 100

[tool.isort]
profile = "black"

[project]
name = "capsel"
version = "0.1.0"
python_requires = ">=3.10"
```